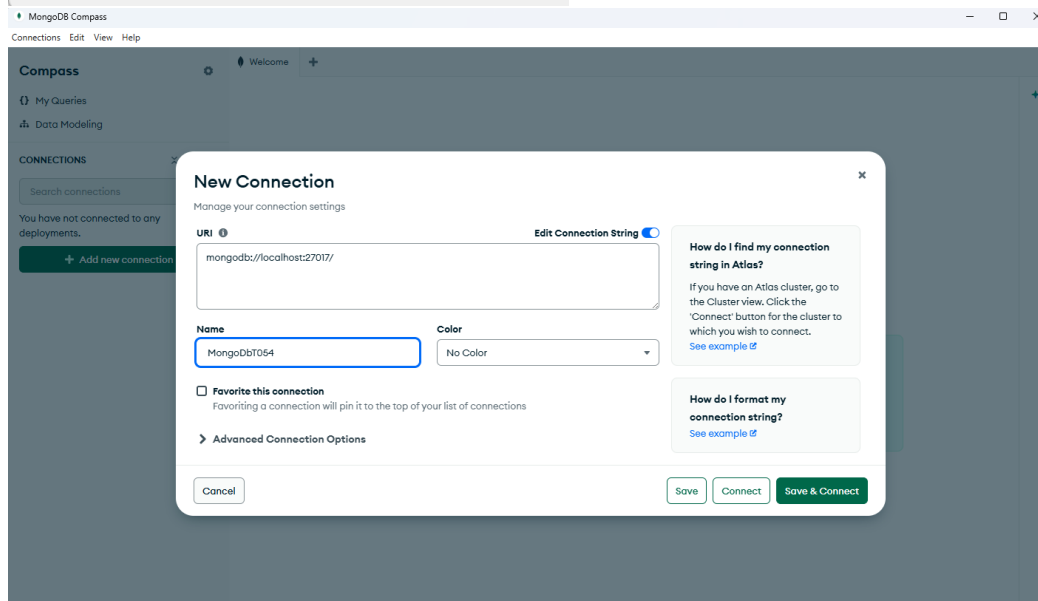
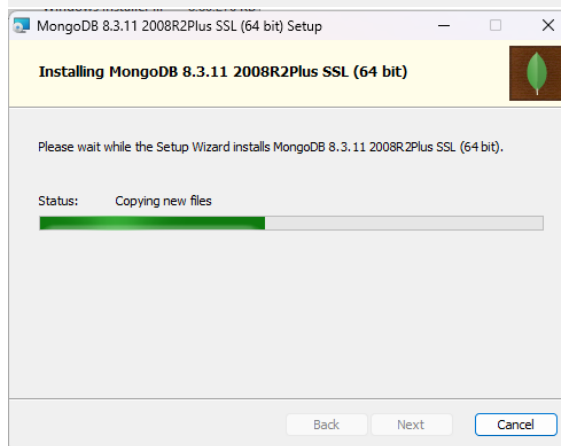
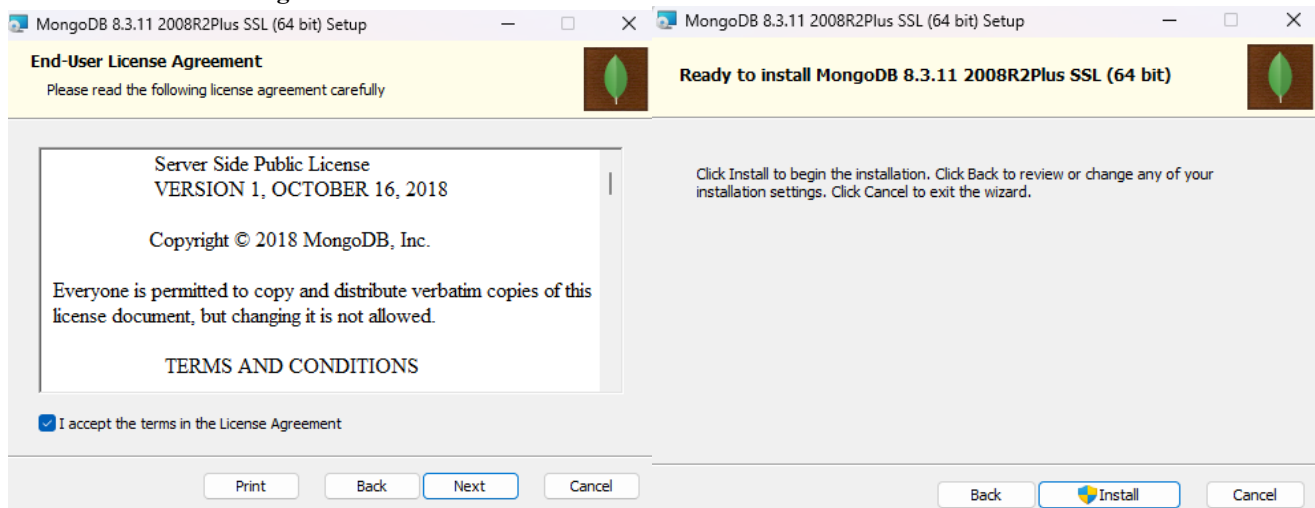


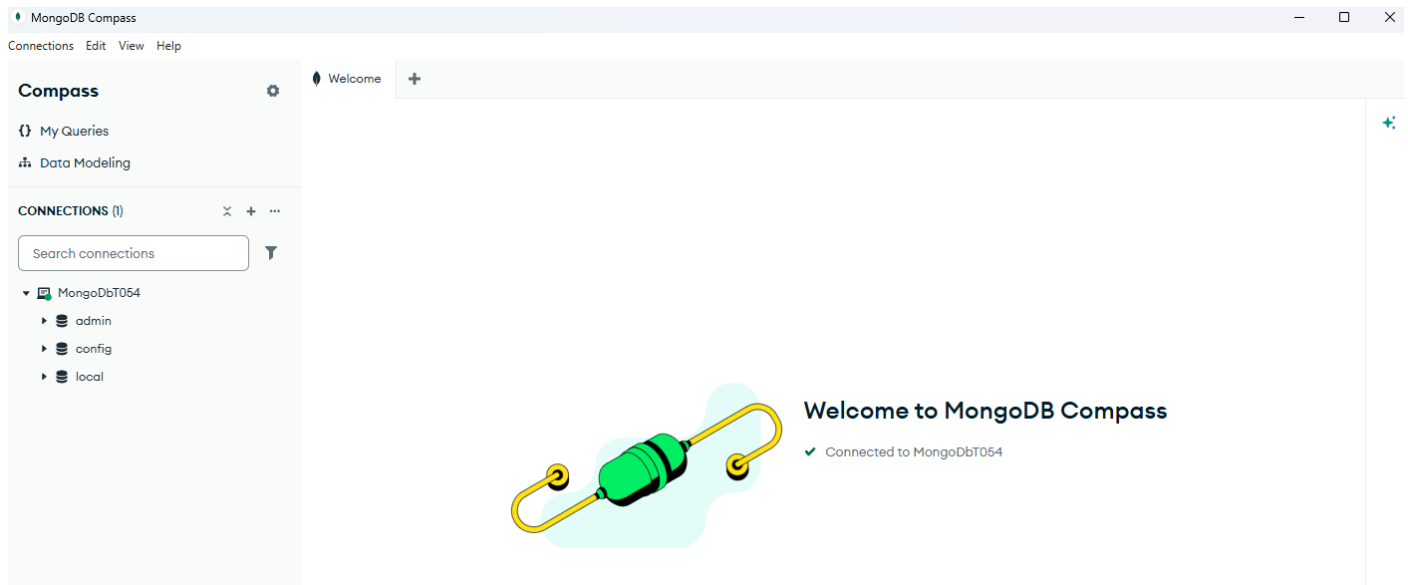
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical
Module - II
Practical – VII

Roll No.:T054	Name:Priti Yadav
Class:TYIT	Batch:2
Date of Assignment: 19/09/26	Date/Time of Submission: 19/09/26

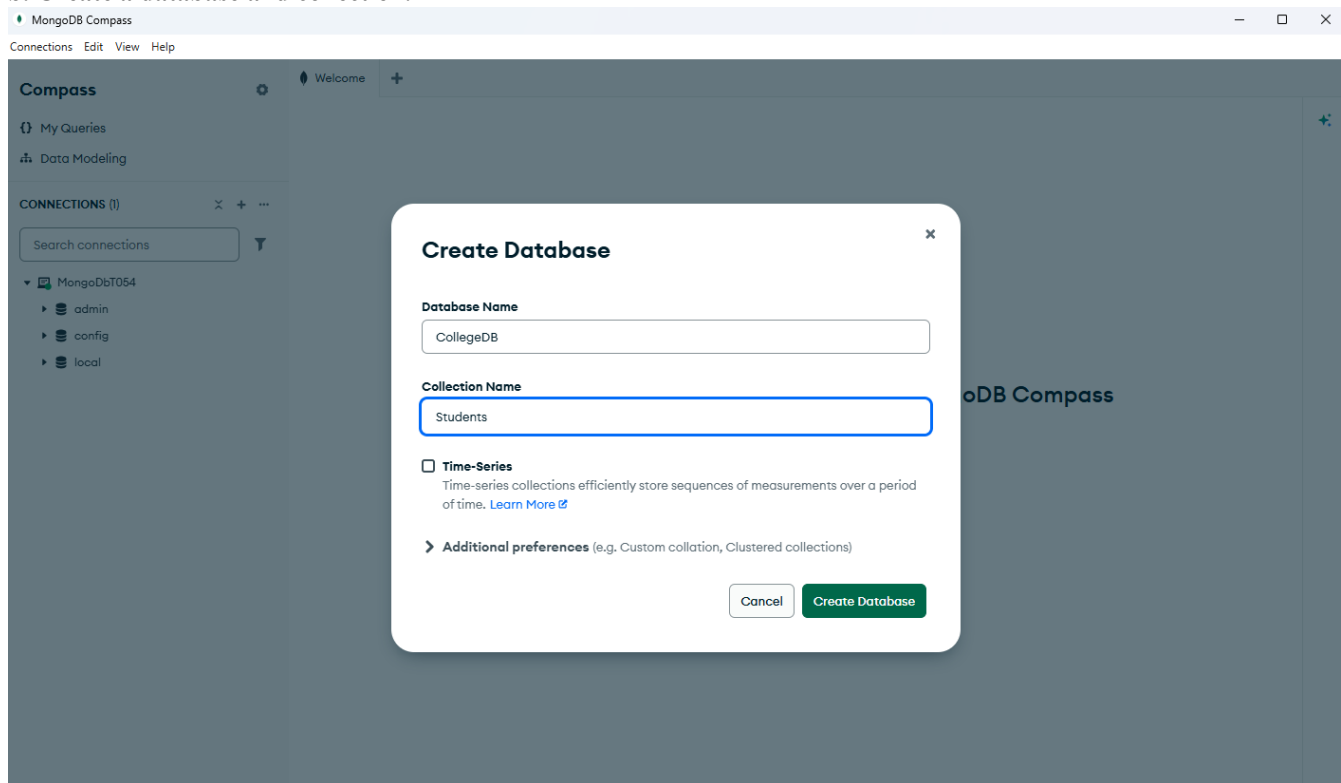
7.MongoDB Basics

a. Install and start MongoDB.

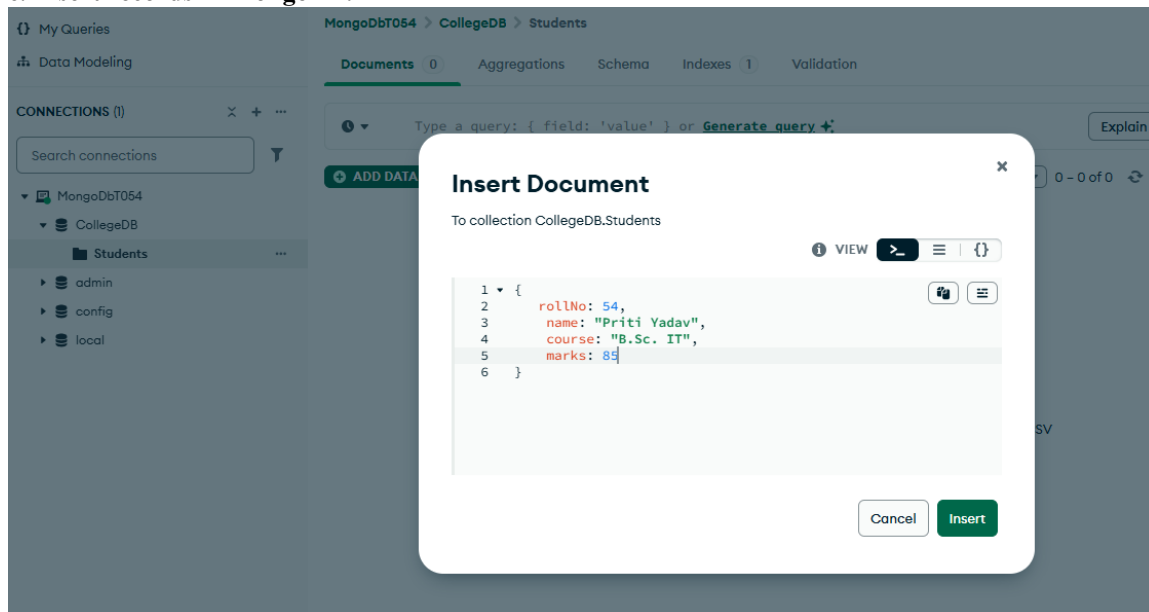


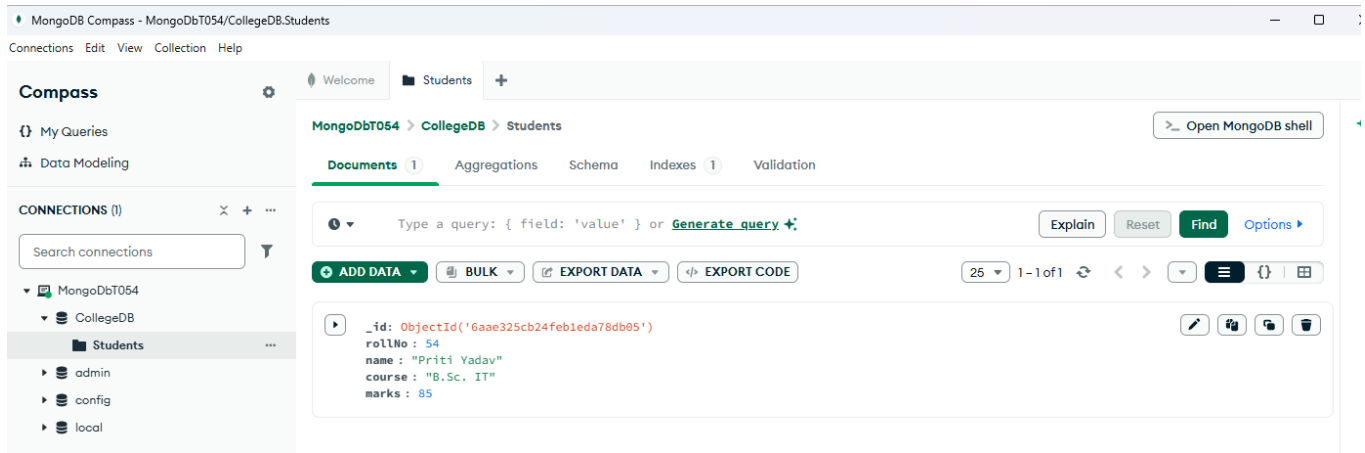


b. Create a database and collection.



c. Insert records in MongoDB.





Open MongoDB shell and select the database: use CollegeDB

```
>_MONGOSH
> use("CollegeDB")
< switched to db CollegeDB
> db["Students"].find()
< {
  _id: ObjectId('6aae325cb24feb1eda78db05'),
  rollNo: 54,
  name: 'Priti Yadav',
  course: 'B.Sc. IT',
  marks: 85
}
```

Insert single record:

```
> db.students.insertOne({
  rollNo: 54,
  name: "Priti Yadav",
  course: "B.Sc. IT",
  marks: 85
})
< {
  acknowledged: true,
  insertedId: ObjectId('6aae3311f6af9dd38e00fd59')
```

Insert Multiple record:

```
> db.students.insertMany([
  {
    rollNo: 55,
    name: "Rahul Sharma",
    course: "B.Sc. IT",
    marks: 85
  },
  {
    rollNo: 56,
    name: "Priya Patil",
    course: "B.Sc. IT",
    marks: 92
  },
  {
    rollNo: 57,
    name: "Amit Joshi",
    course: "B.Sc. CS",
    marks: 78
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aae331ef6af9dd38e00fd5a'),
    '1': ObjectId('6aae331ef6af9dd38e00fd5b'),
    '2': ObjectId('6aae331ef6af9dd38e00fd5c')
  }
}
CollegeDB> |
```

d. View records from the collection.

Command: `db.students.find()`

Command: `db.students.find().pretty()`

```
< {
  _id: ObjectId('6aae3311f6af9dd38e00fd59'),
  rollNo: 54,
  name: 'Priti Yadav',
  course: 'B.Sc. IT',
  marks: 85
}
{
  _id: ObjectId('6aae331ef6af9dd38e00fd5a'),
  rollNo: 55,
  name: 'Rahul Sharma',
  course: 'B.Sc. IT',
  marks: 85
}
```

```
{
  _id: ObjectId('6aae331ef6af9dd38e00fd5b'),
  rollNo: 56,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}
{
  _id: ObjectId('6aae331ef6af9dd38e00fd5c'),
  rollNo: 57,
  name: 'Amit Joshi',
  course: 'B.Sc. CS',
  marks: 78
}
```

e. Update and delete records.

Update one:

```
> db.students.updateOne(
  { rollNo: 54 },
  { $set: { marks: 90 } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
> db.students.find({ rollNo: 54 })
< {
  _id: ObjectId('6aae3311f6af9dd38e00fd59'),
  rollNo: 54,
  name: 'Priti Yadav',
  course: 'B.Sc. IT',
  marks: 90
}
```

```
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

Update Many:

```
> db.students.updateMany(
  { course: "B.Sc. IT" },
  { $set: { course: "Information Technology" } }
)
```

```
> db.students.find()
< {
  _id: ObjectId('6aae3311f6af9dd38e00fd59'),
  rollNo: 54,
  name: 'Priti Yadav',
  course: 'Information Technology',
  marks: 90
}
{
  _id: ObjectId('6aae331ef6af9dd38e00fd5a'),
  rollNo: 55,
  name: 'Rahul Sharma',
  course: 'Information Technology',
  marks: 85
}
```

```
{
  _id: ObjectId('6aae331ef6af9dd38e00fd5b'),
  rollNo: 56,
  name: 'Priya Patil',
  course: 'Information Technology',
  marks: 92
}
```

Delete one:

```
> db.students.deleteOne({
  rollNo: 57
})
< {
  acknowledged: true,
  deletedCount: 1
}
```

```
{
  _id: ObjectId('6aae331ef6af9dd38e00fd5b'),
  rollNo: 56,
  name: 'Priya Patil',
  course: 'B.Sc. IT',
  marks: 92
}
```

```
> db.students.find()
< {
  _id: ObjectId('6aae3311f6af9dd38e00fd59'),
  rollNo: 54,
  name: 'Priti Yadav',
  course: 'B.Sc. IT',
  marks: 90
}
{
  _id: ObjectId('6aae331ef6af9dd38e00fd5a'),
  rollNo: 55,
  name: 'Rahul Sharma',
  course: 'B.Sc. IT',
  marks: 85
}
```

f. Connect MongoDB with Node.js.

MongoDBPractical > JS server.js > ...

```
1  const mongoose = require("mongoose");
2
3  mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
4    .then(() => {
5      console.log("MongoDB Connected Successfully");
6    })
7    .catch((error) => {
8      console.log("MongoDB Connection Failed");
9      console.log(error);
10   });
```

PS C:\Users\mvluc\Desktop\MERN PRAC\MongoDBPractical> node server.js
MongoDB Connected Successfully

MongoDBPractical > JS server.js > ...

```
1  const mongoose = require("mongoose");
2
3  mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
4    .then(async () => {
5      console.log("MongoDB Connected Successfully");
6
7      const students = await mongoose.connection.db
8        .collection("students")
9        .find()
10       .toArray();
11
12      console.log(students);
13    })
14    .catch((error) => {
15      console.log("MongoDB Connection Failed");
16      console.log(error);
17    });
```

PS C:\Users\mvluc\Desktop\MERN PRAC\MongoDBPractical> node server.js
MongoDB Connected Successfully

```
[
  {
    _id: new ObjectId('6aae3311f6af9dd38e00fd59'),
    rollNo: 54,
    name: 'Priti Yadav',
    course: 'Information Technology',
    marks: 90
  },
  {
    _id: new ObjectId('6aae331ef6af9dd38e00fd5a'),
    rollNo: 55,
    name: 'Rahul Sharma',
    course: 'Information Technology',
    marks: 85
  },
  {
    _id: new ObjectId('6aae331ef6af9dd38e00fd5b'),
    rollNo: 56,
    name: 'Priya Patil',
    course: 'Information Technology',
    marks: 92
  }
]
```